

Journal Scope Drift Report

CWTS Leiden citation clustering · macro level · baseline 2020 → 2026 · run 20260721_122750

38,222 Papers analysed	7.9% Overall out-of-scope	3 / 8 High-drift journals	0.289 Mean JSD 2026
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Summary

This report looks at 8 Frontiers journals and asks a simple question: is the type of research each one publishes drifting away from where it started in 2020? Across all of them, about 1 in 13 papers (7.9%) now falls outside the journal's core subject areas. Three journals — Materials, Neurorobotics and Environmental Science — have moved the furthest from their original focus. A page of plain-language commentary and supporting detail follows for each journal.

Scope Drift & OOS Summary by Journal

Journal	Papers	OOS	OOS %	JSD 2026	Drift
Fr. Neurorobotics	1,037	216	20.8%	0.378	High
Fr. Robotics and AI	1,791	204	11.4%	0.314	Moderate
Fr. Chemistry	6,003	649	10.8%	0.239	Low
Fr. Environmental Science	7,093	660	9.3%	0.343	High
Fr. Materials	3,123	246	7.9%	0.379	High
Fr. Surgery	5,885	404	6.9%	0.232	Low
Fr. Aging Neuroscience	5,420	271	5.0%	0.145	Low
Fr. Earth Science	7,870	370	4.7%	0.280	Moderate

A note on how to read this report: This analysis is based purely on citation patterns (which papers cite which), not on reading the papers themselves. Citation-based methods like this can misclassify certain paper types — methods papers and meta-analyses in particular tend to cite (and be cited by) a much wider range of fields than their own subject matter would suggest, so they can appear more “out-of-scope” than they really are. Scope itself is also defined statistically here: a journal's “primary” clusters are simply whichever citation communities together account for 80% of its papers. This threshold is applied uniformly across all journals and hasn't been tuned per journal, so some journals may need a different cutoff to reflect their actual editorial scope. Treat the figures below as a useful signal for further investigation, not a definitive scope audit.

Frontiers in Materials

High

Materials shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

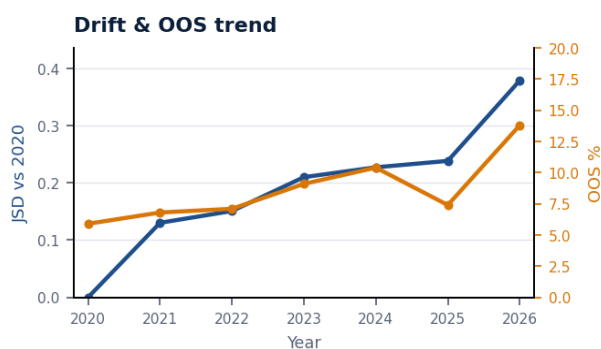
Right now, about 8% of Materials's papers (246 out of 3,123) fall outside its core subject clusters — mostly work related to Comprehensive Dentistry, General Chemistry. This out-of-scope share has risen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is "Sustainable Materials." Its primary clusters have changed since 2020. Areas that have left the core include Comprehensive Materials Science. Newer core areas include Comprehensive Dentistry. In 2026 the largest core topics are Sustainable Materials, Manufacturing And Engineering, and Catalytic Materials (versus Catalytic Materials, Manufacturing And Engineering, and Sustainable Materials in 2020).

Scope split

92% in-scope

8% OOS

3,123 Papers	246 (7.9%) Out-of-scope	0.379 JSD 2026	-0.228 Entropy Δ	5 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Sustainable Materials	In-scope	875	28.0%
Manufacturing And Engineering	In-scope	660	21.1%
Catalytic Materials	In-scope	567	18.2%
Comprehensive Materials Science	In-scope	355	11.4%
Oncology	In-scope	213	6.8%
Energy Technologies	In-scope	100	3.2%

Example papers

In-scope

- An in-depth study on tribological behaviour of polymers and polymer composites: state-of-the-art
- Investigation of rutting deformation mechanisms and high-temperature deformation resistance of asphalt mixtures
- Research on factors influencing high temperature deformation of asphalt mixture and prediction modeling
- Effect of fly ash and metakaolin on early-age autogenous shrinkage and mechanical properties of ultra-high-performance concrete
- Mechanically stimulated self-powered electrochemical sensors: principles, classifications, and future directions

Out-of-scope

- A strategy for fabricating a multifunctional hydrogel with enhanced adhesion and ROS-scavenging capabilities
- Chairside finishing and polishing of modern dental ceramics: material-specific variations in surface roughness
- Enhancement of zirconia–veneering porcelain bond strength via thermal phosphoric acid etching combined with lithium Silicate Coating and the influence of multiple sintering cycles
- Surface roughness and preload loss of implant-supported screw retained three-unit framework fabricated by different techniques
- Intelligent optimisation design of hydraulic presses for composite material forming based on the synergy of an enhanced particle swarm fuzzy control algorithm and an adaptive robust controller

Frontiers in Neurorobotics

High

Neurorobotics shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.38) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

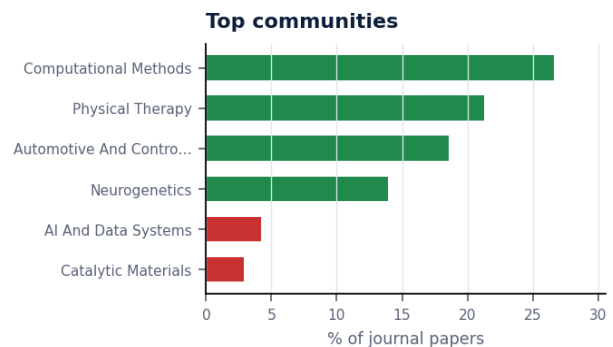
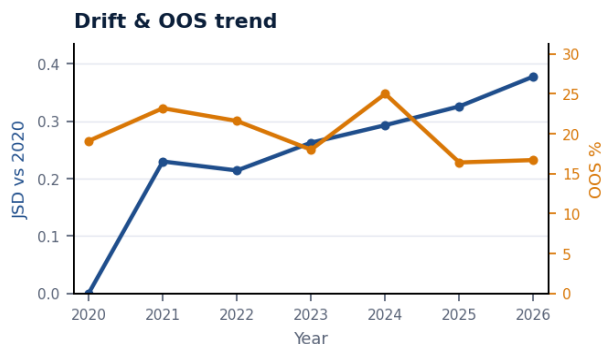
Right now, about 21% of Neurorobotics's papers (216 out of 1,037) fall outside its core subject clusters — mostly work related to AI And Data Systems, Catalytic Materials. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is “Computational Methods.” Its primary clusters have not meaningfully changed since 2020 — still centred on Physical Therapy, Neurogenetics, and Automotive And Control Engineering.

Scope split

79% in-scope

21% OOS

1,037 Papers	216 (20.8%) Out-of-scope	0.378 JSD 2026	-0.4157 Entropy Δ	4 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Computational Methods	In-scope	276	26.6%
Physical Therapy	In-scope	221	21.3%
Automotive And Control Engineering	In-scope	193	18.6%
Neurogenetics	In-scope	144	13.9%
AI And Data Systems	Out-of-scope	44	4.2%
Catalytic Materials	Out-of-scope	30	2.9%

Example papers

In-scope

- Research on embodied agent multimodal perception and real-time path planning algorithms for complex unstructured environments
- Transformer-based human-motion forecasting coupled with safe reinforcement learning for telepresence robot co-navigation
- Emotion estimation from video footage with LSTM
- Interpretable side-aware kinematic-sEMG gait-state representations relevant to adaptive neurorobotic assistance after stroke: a public-dataset study
- Interpretable vision mamba for SAR images classification

Out-of-scope

- Innovative approach of nonlinear controllers design for prosthetic knee performance
- Motion feature extraction based on semi-supervised learning and long short-term memory network in digital dance
- Design an anthropomorphic dexterous hand for expressive piano performance
- A lightweight Chinese–English translation model integrating compressed BERT attention and phrase discard mechanism
- Human-inspired sensorimotor controller for dynamic motion adaptation: a study in robotic arms

Frontiers in Environmental Science

High

Environmental Science shows the clearest signs of scope movement in this review. Compared with where the journal stood in 2020, the mix of research it publishes has shifted noticeably — a change of this size (JSD 0.34) means the papers coming in today look meaningfully different from the ones that defined the journal a few years ago.

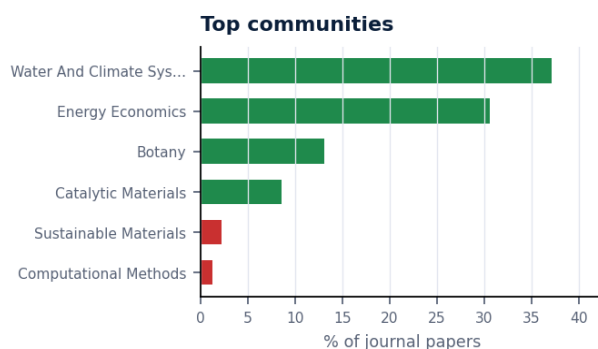
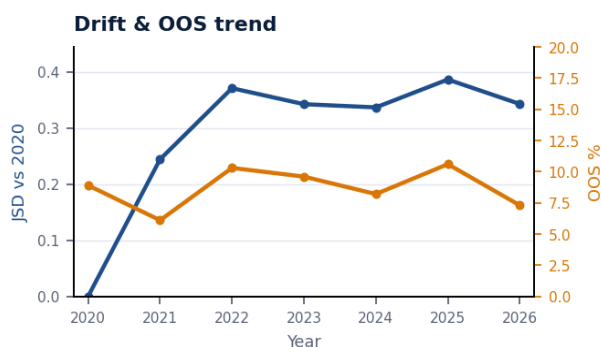
Right now, about 9% of Environmental Science's papers (660 out of 7,093) fall outside its core subject clusters — mostly work related to Sustainable Materials, Computational Methods. This out-of-scope share has fallen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is “Water And Climate Systems.” Its primary clusters have changed since 2020. Areas that have left the core include Catalytic Materials. Newer core areas include Energy Economics. In 2026 the largest core topics are Water And Climate Systems, Energy Economics, and Botany (versus Water And Climate Systems, Catalytic Materials, and Botany in 2020).

Scope split

91% in-scope

9% OOS

7,093 Papers	660 (9.3%) Out-of-scope	0.343 JSD 2026	-0.0436 Entropy Δ	3 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Water And Climate Systems	In-scope	2,629	37.1%
Energy Economics	In-scope	2,171	30.6%
Botany	In-scope	926	13.1%
Catalytic Materials	In-scope	608	8.6%
Sustainable Materials	Out-of-scope	156	2.2%
Computational Methods	Out-of-scope	91	1.3%

Example papers

In-scope

- Spatial distribution characteristics and optimization strategies of carbon sequestration service in the yellow river Basin ecosystems: a case study of Shanxi section
- A 20-year spatio-temporal analysis of 3D radar-based hail tracks in Germany: trends and regional differences
- A novel approach to TDS estimation in desalination systems using differential pressure sensing
- Multi-criteria ranking of renewable energy alternatives for smart grids using expert-weighted TOPSIS
- Effects of the polycyclic aromatic hydrocarbon anthracene on the survival of the semi-aquatic grasshopper *Cornops aquaticum* (Bruner, 1906): an experimental toxicological study

Out-of-scope

- Trends and insights in machine learning-based landslide susceptibility research and early warning practices: a bibliometric analysis
- Citizen science as a supporting mechanism for China's national ecological monitoring system building (2021–2025): advances, challenges, and strategic perspectives
- Fire-YOLO26: an NMS-free dual-stream RGB–thermal network for real-time UAV forest fire detection in smoke-challenging conditions
- Development of a cascade recognition method for short-term heavy precipitation over the Yangtze River Delta
- Spatiotemporal monitoring of grasshopper habitats in China: integrating ensemble machine learning, degree-day modeling, and multi-source remote sensing data

Frontiers in Robotics and AI

Moderate

Robotics and AI is showing a moderate amount of movement away from its 2020 starting point (JSD 0.31). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

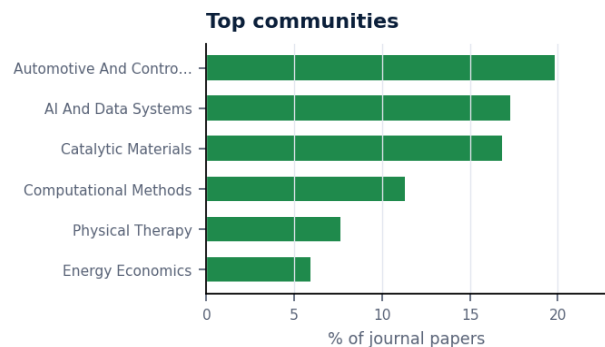
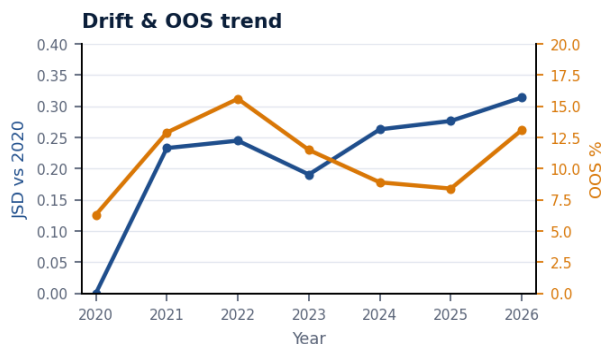
Right now, about 11% of Robotics and AI's papers (204 out of 1,791) fall outside its core subject clusters — mostly work related to Atmospheric Science, Telecommunications Engineering. This out-of-scope share has risen since 2020. The spread of topics it covers has stayed fairly stable. Its largest single topic area today is "Automotive And Control Engineering." Its primary clusters have changed since 2020. Newer core areas include Energy Economics. In 2026 the largest core topics are Automotive And Control Engineering, AI And Data Systems, and Computational Methods (versus Automotive And Control Engineering, Catalytic Materials, and Physical Therapy in 2020).

Scope split

89% in-scope

11% OOS

1,791 Papers	204 (11.4%) Out-of-scope	0.314 JSD 2026	+0.0379 Entropy Δ	7 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Automotive And Control Engineering	In-scope	354	19.8%
AI And Data Systems	In-scope	309	17.3%
Catalytic Materials	In-scope	301	16.8%
Computational Methods	In-scope	203	11.3%
Physical Therapy	In-scope	136	7.6%
Energy Economics	In-scope	106	5.9%

Example papers

In-scope

- Social robots: a meta-analysis of learning outcomes
- Modelling C-arm fluoroscopy and operating table kinematics via machine learning
- SANE (eaSy gAit aNalysis systEm): real-time dual-depth, marker-less system characterisation and robustness of gait parameters
- Multi-strategy Sea Horse Optimization algorithm for UAV path planning
- RAISE-FER: a massive cross-dataset augmented facial expression dataset

Out-of-scope

- Automating PINN-based kinematic resolution of robotic joints using robotic process automation frameworks
- Data-driven analysis of Armeo Spring performance across neurological disorders: implications for personalized upper limb neurorehabilitation
- Automated capture and transfer of human facial expressions to humanoid robots for realistic patient simulation
- Optimizing avian flight dynamics with a synergetic bio-inspired and machine learning approach
- Dynamic variance-aware federated tuning for efficient autonomous vehicle perception under non-IID settings

Frontiers in Earth Science

Moderate

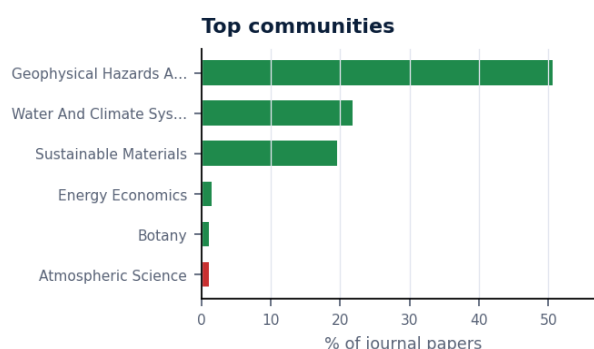
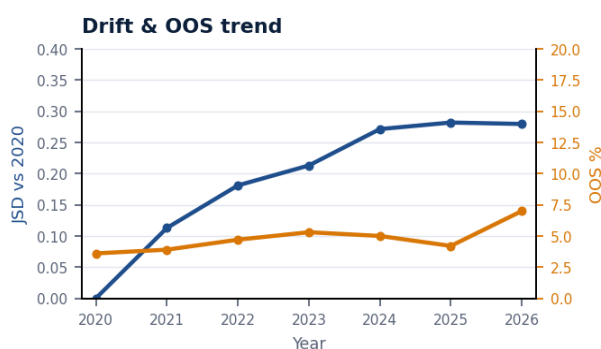
Earth Science is showing a moderate amount of movement away from its 2020 starting point (JSD 0.28). This isn't dramatic, but the journal's content mix has evolved enough to be worth keeping an eye on.

Right now, about 5% of Earth Science's papers (370 out of 7,870) fall outside its core subject clusters — mostly work related to Atmospheric Science, Computational Methods. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is “Geophysical Hazards And Engineering.” Its primary clusters have changed since 2020. Newer core areas include Sustainable Materials. In 2026 the largest core topics are Geophysical Hazards And Engineering, Sustainable Materials, and Water And Climate Systems (versus Geophysical Hazards And Engineering and Water And Climate Systems in 2020).

Scope split

95% in-scope

7,870 Papers	370 (4.7%) Out-of-scope	0.280 JSD 2026	+0.1262 Entropy Δ	3 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Geophysical Hazards And Engineering	In-scope	3,981	50.6%
Water And Climate Systems	In-scope	1,716	21.8%
Sustainable Materials	In-scope	1,534	19.5%
Energy Economics	In-scope	109	1.4%
Botany	In-scope	87	1.1%
Atmospheric Science	Out-of-scope	80	1.0%

Example papers

In-scope

- A novel sedimentologically controlled accumulation pattern for marine shale gas: insights from the “source-reservoir-seal” configurations of lower Carboniferous shale in the Yaziluo rift trough, Southern China
- Machine learning–based flood inundation mapping using fused optical and SAR remote sensing: a case study of the Miyun flood
- Provenance and tectonic setting of the Lower Jurassic-Lower Cretaceous shales from the northeastern Iraq: mineralogical and geochemical approaches
- Multiscale characterization of pore–throat structures and flow mechanisms in low-permeability reservoirs
- Mapping the impacts of recurrent floods and windstorms in Bauchi, Nigeria: a hybrid multi-criteria approach using fuzzy-AHP

Out-of-scope

- Study on the dynamic response characteristics of hydraulic support columns under multi-condition impact loads
- MVarGOSIM: an MPS algorithm for characterizing complex structures with multiple variables
- Beyond structured knowledge: performance boundaries of ChatGPT in geological-hazard question answering and the need for human-in-the-loop oversight
- Comprehensive evaluation and zoning study of potential geothermal water resources in Qinghai Province, northeastern margin of the Tibetan Plateau
- Semantic rule-guided three-dimensional reservoir modeling method using an improved multiple-point geostatistics simulation

Frontiers in Chemistry

Low

Chemistry remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.24), which suggests the journal is publishing consistently within its intended focus.

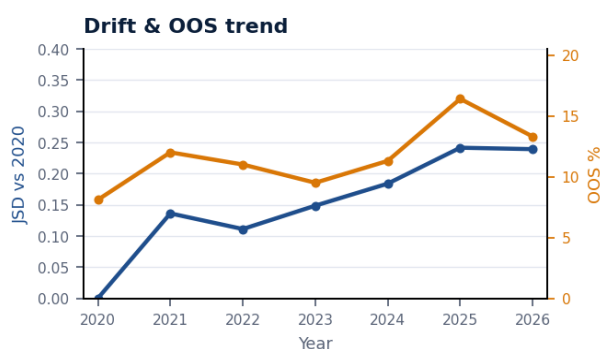
Right now, about 11% of Chemistry's papers (649 out of 6,003) fall outside its core subject clusters — mostly work related to Immunology And Disease, Microbial Diversity And Interactions. This out-of-scope share has risen since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is "Catalytic Materials." Its primary clusters have changed since 2020. Areas that have left the core include Comprehensive Materials Science. Newer core areas include Natural Products And Herbal Medicine and Machine Learning. In 2026 the largest core topics are Catalytic Materials, Oncology, and General Chemistry (versus Catalytic Materials, General Chemistry, and Oncology in 2020).

Scope split

89% in-scope

11% OOS

6,003 Papers	649 (10.8%) Out-of-scope	0.239 JSD 2026	+0.2844 Entropy Δ	5 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Catalytic Materials	In-scope	2,051	34.2%
Oncology	In-scope	1,068	17.8%
General Chemistry	In-scope	1,067	17.8%
Comprehensive Materials Science	In-scope	434	7.2%
Natural Products And Herbal Medicine	In-scope	430	7.2%
Botany	In-scope	123	2.0%

Example papers

In-scope

- HIT101308137 and HIT104293658 nominate dual target chemotypes for PTPN1 and PTPN2 with preliminary selectivity in colorectal cancer cells
- Editorial: Exploration of the role of heme proteins in biology with experimental and computational methods
- Formation and transfer patterns of key aroma compounds in cigar tobacco based on SBSE-GC-O-MS, pyrolysis-GC-O-MS and flue gas analysis correlation studies
- Diol-enhanced natural deep eutectic solvents for efficient poplar pretreatment
- Influence of substitution patterns on the antimicrobial properties of pyrrole sulfonamide scaffolds

Out-of-scope

- Entropy comparison of benzothiadiazole-based covalent-organic frameworks
- Design, synthesis, and biological evaluation of derivatives from p-toluic hydrazide: lead compound C17 exhibits membrane-disrupting and anti-biofilm activities against Staphylococcus aureus and other gram-positive bacteria
- Lipidated peptides and the limits of chemical control
- Biophoton emission promotes the modern scientific utilization of medicinal plants: from theoretical integration to research applications
- Emerging insights into chemistry and therapeutic potentials of functionalized hexahydroquinolines

Frontiers in Surgery

Low

Surgery remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.23), which suggests the journal is publishing consistently within its intended focus.

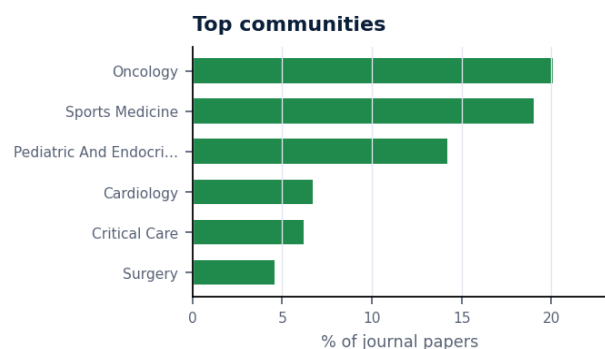
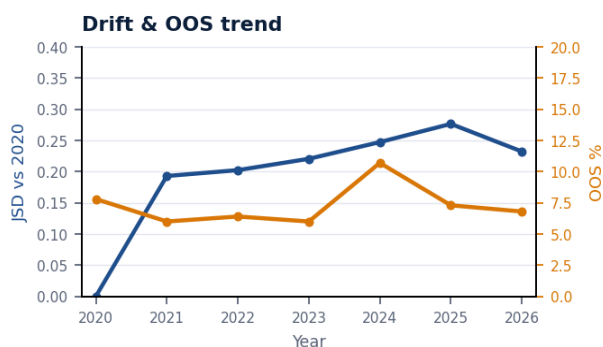
Right now, about 7% of Surgery's papers (404 out of 5,885) fall outside its core subject clusters — mostly work related to peripheral topics. This out-of-scope share has fallen since 2020. At the same time, the journal is concentrating more tightly around a smaller set of themes rather than spreading out. Its largest single topic area today is "Oncology." Its primary clusters have changed since 2020. Areas that have left the core include Immunology And Disease and Audiology And Speech Therapy. Newer core areas include Neurogenetics and Endocrinology And Diabetes. In 2026 the largest core topics are Sports Medicine, Oncology, and Pediatric And Endocrine Care (versus Oncology, Pediatric And Endocrine Care, and Sports Medicine in 2020).

Scope split

93% in-scope

7% OOS

5,885 Papers	404 (6.9%) Out-of-scope	0.232 JSD 2026	-0.1047 Entropy Δ	10 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Oncology	In-scope	1,181	20.1%
Sports Medicine	In-scope	1,119	19.0%
Pediatric And Endocrine Care	In-scope	834	14.2%
Cardiology	In-scope	397	6.7%
Critical Care	In-scope	362	6.2%
Surgery	In-scope	272	4.6%

Example papers

In-scope

- Establishing an implementation framework for clinical nursing guidelines in hepatobiliary and accelerated rehabilitation surgery: based on the Ottawa research application model
- Case Report: Arthroscopic synovectomy and loose body removal for charcot knee in an adolescent with congenital insensitivity to pain with anhidrosis
- Comparative outcomes of natural orifice specimen extraction surgery versus totally laparoscopic surgery for right-sided colon cancer: a single-centre propensity score-matched study
- Outcomes of malignant orbital tumors: a 22-year Norwegian single-center series
- Rewiring faces: advances and outcomes in facial nerve reconstruction after facial vascularized composite allotransplantation

Out-of-scope

- Case Report: Report of 2 cases of periprosthetic joint infection caused by Brucella after joint replacement and literature review
- Comparison of short-term outcomes of open, laparoscopic, and robotic surgery for Kasai portoenterostomy in biliary atresia: a 10-year single center study
- Effects of obesity on post-surgical recovery and functional outcomes in rotator cuff tear repair
- Case Report: Unclassifiable cerebellar high-grade neuroepithelial tumor with a CCDC6::RET fusion manifesting explosive recurrence
- Bibliometric and network analysis of acute conus medullaris syndrome and conus-related complications after lumbar spine procedures

Frontiers in Aging Neuroscience

Low

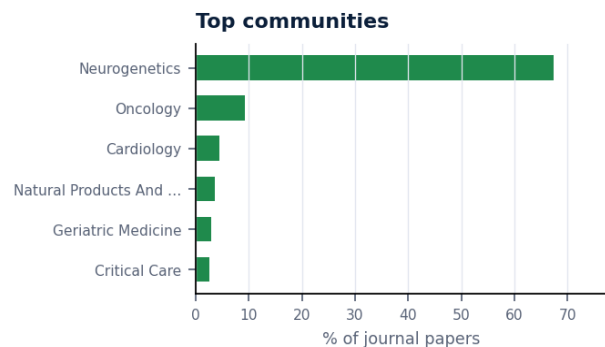
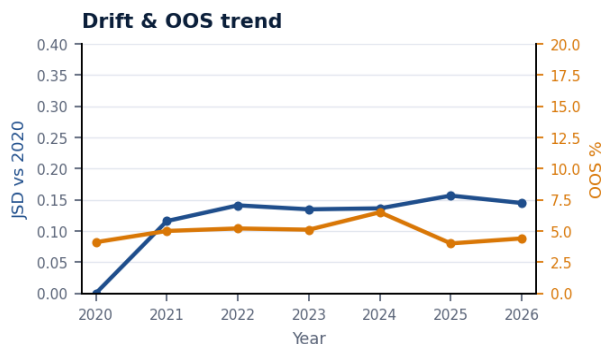
Aging Neuroscience remains close to its original scope. Its content mix has changed only slightly since 2020 (JSD 0.14), which suggests the journal is publishing consistently within its intended focus.

Right now, about 5% of Aging Neuroscience's papers (271 out of 5,420) fall outside its core subject clusters — mostly work related to Ophthalmology, Paediatrics And Adolescent Medicine. This out-of-scope share has stayed similar since 2020. The journal is also spreading across a wider range of topics rather than staying tightly focused. Its largest single topic area today is "Neurogenetics." Its primary clusters have changed since 2020. Newer core areas include Cardiology. In 2026 the largest core topics are Neurogenetics, Oncology, and Cardiology (versus Neurogenetics and Oncology in 2020).

Scope split

95% in-scope

5,420 Papers	271 (5.0%) Out-of-scope	0.145 JSD 2026	+0.1983 Entropy Δ	3 Primary clusters	80.0% Primary coverage
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Community composition

Community	Status	Papers	Share of journal
Neurogenetics	In-scope	3,652	67.4%
Oncology	In-scope	503	9.3%
Cardiology	In-scope	244	4.5%
Natural Products And Herbal Medicine	In-scope	195	3.6%
Geriatric Medicine	In-scope	163	3.0%
Critical Care	In-scope	143	2.6%

Example papers

In-scope

- Identification and validation of an interpretable EEG-based machine learning model for the diagnosis of post-stroke cognitive impairment
- Neural and motor mechanisms of handwriting: from healthy aging to neurodegenerative disorders
- Neuromelanin, iron and MRI measurements in midbrain tissues of Parkinson's and Alzheimer's subjects
- Age-dependent vulnerability to spatial memory interference in APP/PS1 mice
- Mechanisms to medicines: navigating drug repurposing strategies in Alzheimer's disease

Out-of-scope

- The aggregate proteome of *Caenorhabditis elegans* mitochondria implicates shared mechanisms of aging and Alzheimer's disease
- Effect of deep brain stimulation on dysphagia in Parkinson's disease: mechanisms, evidence, and outlook
- Which aspects of the everyday behavior of older dogs correlate with performance on a visuo-spatial memory test and the canine cognitive dysfunction rating scale (CCDR)?
- A transcriptomic resource for glial GABA-associated ASH neuronal aging and candidate pathways
- Alterations in energy production in a *Drosophila* model for the X-linked dystonia-parkinsonism-related *Taf1* deficiency